Probability, Statistics and Random Processes

Third Edition

T Veerarajan
Professor of Mathematics and Head
Science and Humanities Department
Sree Sowdambika College of Engineering
Aruppukottai, Tamil Nadu

Tata McGraw-Hill Publishing Company Limited
NEW DELHI

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Tata McGraw-Hill

Published by the Tata McGraw-Hill Publishing Company Limited,
7 West Patel Nagar, New Delhi 110 008.

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This edition can be exported from India only by the publishers,
Tata McGraw-Hill Publishing Company Limited.

ISBN 13: 978-0-07-066925-3
ISBN 10: 0-07-066925-2

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Published by the Tata McGraw-Hill Publishing Company Limited, 7 West Patel Nagar, New Delhi 110 008, typeset at Bukprint India, B-180A, Guru Nanak Pura, Laxmi Nagar-110 092 and printed at Adarsh Printers, C-50-51, Mohan Park, Naveen Shahdara, Delhi 110 032

Cover: SDR Printers

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REGRESSION AND CORRELATION

Regression

A **regression line** is a straight line that describes how a response variable Y changes as an explanatory variable X changes. Regression lines are commonly used to **predict values**.

Equation of the Regression Line of Y on X

By the **principle of least squares**, the normal equations used to determine the values of a and b are:

$$\sum y_i = a \sum x_i + nb$$

$$\sum x_i y_i = a \sum x_i^2 + b \sum x_i$$

Dividing the first equation by n:

$$\bar{y} = a\bar{x} + b$$

Hence, the regression equation of Y on X is:

$$y - \bar{y} = \frac{\rho_{xy}}{\sigma_x^2}(x - \bar{x})$$

where

$$\rho_{xy} = cov(X, Y)$$

Alternate Form

$$y - \bar{y} = r_{xy}\frac{\sigma_y}{\sigma_x}(x - \bar{x})$$

where

$$r_{xy} = \frac{\rho_{xy}}{\sigma_x \sigma_y}$$

Regression Line of X on Y

Similarly, the regression equation of X on Y is:

$$x - \bar{x} = r_{xy}\frac{\sigma_x}{\sigma_y}(y - \bar{y})$$

or equivalently

$$x - \bar{x} = \frac{\rho_{xy}}{\sigma_y^2}(y - \bar{y})$$

Problem 1

For the following data, find the regression line of Y on X.

X	1	2	3	4	5	8	10
Y	9	8	10	12	14	16	15

Solution

$$\bar{x} = \frac{\sum x_i}{n} = \frac{33}{7} = 4.714$$

$$\bar{y} = \frac{\sum y_i}{n} = \frac{84}{7} = 12$$

Construct the following table:

X	Y	XY	X²
1	9	9	1
2	8	16	4
3	10	30	9
4	12	48	16
5	14	70	25
8	16	128	64
10	15	150	100

The regression coefficient:

$$b_{yx} = \frac{n\sum XY - \sum X \sum Y}{n\sum X^2 - (\sum X)^2}$$

$$b_{yx} = 0.867$$

Regression equation:

$$y - \bar{y} = b_{yx}(x - \bar{x})$$

$$y - 12 = 0.867(x - 4.714)$$

$$y = 0.867x + 7.9129$$

Problem 2

From the following data, fit two regression equations using the **actual mean method**.

X	1	2	3	4	5	6	7
Y	2	4	7	6	5	6	5

Solution

$$\bar{x} = \frac{28}{7} = 4$$

$$\bar{y} = \frac{35}{7} = 5$$

Construct the table:

X	Y	$X-\bar{x}$	$Y-\bar{y}$	X^2	Y^2	XY
1	2	-3	-3	9	9	9
2	4	-2	-1	4	1	2
3	7	-1	2	1	4	-2
4	6	0	1	0	1	0
5	5	1	0	1	0	0
6	6	2	1	4	1	2
7	5	3	0	9	0	0

$$\sum X^2 = 28$$

$$\sum Y^2 = 16$$

$$\sum XY = 11$$

Regression coefficients:

$$b_{yx} = \frac{\sum XY}{\sum X^2} = \frac{11}{28} = 0.3928$$

$$b_{xy} = \frac{\sum XY}{\sum Y^2} = \frac{11}{16} = 0.6875$$

Regression equation of Y on X

$$y - 5 = 0.3928(x - 4)$$

$$y = 0.393x + 3.428$$

Regression equation of X on Y

$$x - 4 = 0.6875(y - 5)$$

$$x = 0.688y + 0.56$$

Problem 3

From the following results obtain the regression equations and estimate crop yield when rainfall is **29 cm**, and rainfall when yield is **600 kg**.

	Yield Y (kg)	Rainfall X (cm)
Mean	508.4	26.7
S.D.	36.8	4.6

Correlation coefficient

$$\rho = 0.52$$

Solution

$$\bar{x} = 26.7$$

$$\bar{y} = 508.4$$

$$\sigma_x = 4.6$$

$$\sigma_y = 36.8$$

Regression coefficients:

$$b_{yx} = \rho \frac{\sigma_y}{\sigma_x}$$

$$b_{xy} = \rho \frac{\sigma_x}{\sigma_y}$$

$$b_{yx} = 4.16$$

$$b_{xy} = 0.065$$

Regression Equations

$$y = 4.16x + 397.328$$

$$x = 0.065y - 6.346$$

Estimation

When $X = 29$

$$y = (4.16)(29) + 397.328$$

$$y = 517.968 \text{ kg}$$

When $Y = 600$

$$x = (0.065)(600) - 6.346$$

$$x = 32.654 \text{ cm}$$

Exercise Problem

Find regression equations for the data and estimate:

- Y when $X = 71$
- X when $Y = 70$

X	65	66	67	67	68	69	70	72
Y	67	68	65	68	72	72	69	71

Multiple Linear Regression

If the number of independent variables in a regression model is **more than one**, the model is called **multiple regression**.

Many real-world applications require multiple regression models.

Regression Model with Two Independent Variables

Suppose two independent variables exist.

$$Y = b_0 + b_1 X_1 + b_2 X_2$$

Normal Equations

$$\sum Y = nb_0 + b_1 \sum X_1 + b_2 \sum X_2$$

$$\sum X_1 Y = b_0 \sum X_1 + b_1 \sum X_1^2 + b_2 \sum X_1 X_2$$

$$\sum X_2 Y = b_0 \sum X_2 + b_1 \sum X_1 X_2 + b_2 \sum X_2^2$$

Problem (Multiple Regression)

Annual sales revenue depends on:

- X_1: Sales force
- X_2: Advertising expenditure

Data

Annual Sales Y	20	23	25	27	21	29	22	24	27	35
Sales Force X1	8	13	8	18	23	16	10	12	14	20
Advertising X2	28	23	38	16	20	28	23	30	26	32

Regression model:

$$Y = b_0 + b_1 X_1 + b_2 X_2$$

Calculation Table

Y	X1	X2	X1²	X2²	X1X2	YX1	YX2
20	8	28	64	784	224	160	560
23	13	23	169	529	299	299	529
25	8	38	64	1444	304	200	950
27	18	16	324	256	288	486	432
21	23	20	529	400	460	483	420
29	16	28	256	784	448	464	812
22	10	23	100	529	230	220	506
24	12	30	144	900	360	288	720
27	14	26	196	676	364	378	702
35	20	32	400	1024	640	700	1120

Totals:

$$\sum Y = 253$$

$$\sum X_1 = 142$$

$$\sum X_2 = 264$$

$$\sum X_1^2 = 2246$$

$$\sum X_2^2 = 7326$$

$$\sum X_1 X_2 = 3617$$

$$\sum X_1 Y = 3678$$

$$\sum X_2 Y = 6751$$

Simultaneous Equations

$$253 = 10b_0 + 142b_1 + 264b_2$$

$$3678 = 142b_0 + 2246b_1 + 3617b_2$$

$$6751 = 264b_0 + 3617b_1 + 7326b_2$$

Solution:

$$b_0 = 5.1483$$

$$b_1 = 0.6190$$

$$b_2 = 0.4304$$

Regression Model

$$Y = 5.1483 + 0.6190X_1 + 0.4304X_2$$

Multiple and Partial Correlation

Multiple Correlation

When one variable is influenced by several other variables, the correlation is called **multiple correlation**.

Multiple Correlation Coefficient

For variables X_1, X_2, X_3

$$R^2_{1.23} = \frac{r^2_{12} + r^2_{13} - 2r_{12}r_{13}r_{23}}{1 - r^2_{23}}$$

Notes:

1. $R^2_{1.23} \leq 1$
2. $0 \leq R_{1.23} \leq 1$

Properties of Multiple Correlation Coefficient

1. It measures the association between observed and predicted values of the dependent variable.
2. The value lies between **0 and 1**.
3. If it equals **1**, the association is perfect.
4. If it equals **0**, the dependent variable is uncorrelated with independent variables.
5. It is always greater than or equal to any total correlation coefficient.
6. The least squares method gives the maximum multiple correlation coefficient.

Partial Correlation

Sometimes correlation between two variables is influenced by a third variable.

When the effect of the third variable is removed, the remaining correlation is called **partial correlation**.

Partial Correlation Formulas

1️⃣ Between X_1 and X_2 controlling X_3

$$r_{12.3} = \frac{r_{12} - r_{13}r_{23}}{\sqrt{(1 - r^2_{13})(1 - r^2_{23})}}$$

2️⃣ Between X_1 and X_3 controlling X_2

$$r_{13.2} = \frac{r_{13} - r_{12}r_{23}}{\sqrt{(1 - r^2_{12})(1 - r^2_{23})}}$$

3️⃣ Between X_2 and X_3 controlling X_1

$$r_{23.1} = \frac{r_{23} - r_{21}r_{31}}{\sqrt{(1 - r^2_{21})(1 - r^2_{31})}}$$

Example 1

Given

$$r_{12} = 0.77$$

$$r_{13} = 0.72$$

$$r_{23} = 0.52$$

Find:

- $r_{12.3}$
- $R_{1.23}$

Partial Correlation

$$r_{12.3} = \frac{0.77 - (0.72)(0.52)}{\sqrt{(1 - 0.72^2)(1 - 0.52^2)}}$$

$$r_{12.3} = 0.67$$

Multiple Correlation

$$R^2_{1.23} = \frac{0.77^2 + 0.72^2 - 2(0.77)(0.72)(0.52)}{1 - 0.52^2}$$

$$R^2_{1.23} = 0.73$$

$$R_{1.23} = \sqrt{0.73} = 0.86$$

Example 2

Given

$$r_{12} = 0.71$$

$$r_{23} = r_{31} = 0.50$$

Partial Correlation

$$r_{23.1} = \frac{0.50 - (0.71)(0.50)}{\sqrt{(1 - 0.71^2)(1 - 0.50^2)}}$$

$$r_{23.1} = 0.24$$

Multiple Correlation

$$R^2_{1.23} = \frac{0.71^2 + 0.50^2 - 2(0.71)(0.50)(0.50)}{1 - 0.50^2}$$

$$R^2_{1.23} = 0.552$$

$$R_{1.23} = \sqrt{0.552} = 0.73$$